

4.5.1 Airport and Runway Data Formats

In general, an airport data record is followed by one or more runway data records. This pattern can be repeated indefinitely until all of the airports are entered. The airport and runway data can be provided in a mix of units of measure and obstacle reference points as specified in the airport data records. STAS will convert the data appropriately and show the information on the output page using a uniform set of units.

There is more than one choice of format for both airport and runway records each having its own advantages and disadvantages. For example, the data in AIRPORT1 format is not sensitive to column position, but each data item must be enclosed with single quotes. Alternatively, AIRPORT2 format can be used. In AIRPORT2 format, quotes are not needed but column position is critical.

AIRPORT1 Format:

Line 1: AIRPORT1

Line 2: The following data items, in order:

Airport code (up to 4 characters enclosed in single quotes)

Airport name (up to 18 characters enclosed in single quotes)

City (up to 18 characters enclosed in single quotes)

Six characters obstacle reference string (enclosed in single quotes)

Defined as follows:

Character 1 - Height units (For M)

Character 2 - Distance units (For M)

Characters 3-4 - Height reference (BRor LOor MS), where

BR= Obstacle heights measured from break release end of runway.

LO= Obstacle heights measured from liftoff end of runway.

MS= Obstacle heights measured from mean sea level. Height will be translated to runway end referencing using MSL ELEV setting and elevation input.

Characters 5-6 - Distance reference (BRor LO) Elevation. Used in airport pressure altitude definition.

Example:

AIRPORT1	'KZEB'	'ZNEW INTL'	'SEATTLE, WA'	'FFBRBR'	250
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Note: If the airport name includes a single quotation mark, use two consecutive quotation marks instead.

For example:

'Chicago-O' 'Hare' can be
used for:
Chicago-O'Hare

AIRPORT2 Format:

Line 1: AIRPORT2
Line 2: The following data items, in order:

- Airport code - Columns 1 - 4
- Airport name - Columns 7 - 24
- City - Columns 27 - 44
- Height units - Column 47 (For M)
- Distance units - Column 48 (For M)
- Height Ref - Columns 49 - 50 (BRor LOor MS)
 - BR = Brake release end of runway.
 - LO = Liftoff end of runway.
 - MS = Mean sea level.
 - Height will be translated to runway end referencing using MSL ELEV setting and elevation input.
- Distance Ref - Columns 51 - 52 (BRor LO)
 - BR = Brake release end of runway.
 - LO = Liftoff end of runway.
- Elevation - Columns 55 - 61
 - Used in airport pressure altitude definition.

Example:

AIRPORT2					
ZXXX	YANKEE DOODLE INTL	SEATTLE, WA	FFBRBR	250	

12345678901234567890123456789012345678901234567890123456789012345678901

10 20 30 40 50 60

⇒ Note: The obstacle reference points for input airport/runway data are controlled by the reference string in the airport line. The output data in the takeoff analysis page is controlled by the input in OBSD(2) per the SCAP specification and series guides/release notes. Regardless of the input string, obstacles output will be referenced from the liftoff end of the runway if OBSD(2) = 1 in your input file. Conversely, an input of OBSD(2) = 0 will result in output references from the brake release end of the runway. STAS defaults OBSD(2) = 1.

The runway data records can be provided in four formats. RWYS and RWYU are designed for straight out departures. Although RWYT and RWYV allow inputs related to turning departures, analysis of turning departures is generally not supported.

RWYS Format: Straight out departures

Line 1: RWYS columns 2 through 5.
 (Optional) Angle of taxiway entry to runway in degrees.
 (Optional) Missed approach gradient requirement in percent.
 To be computed at actual missed approach speed.
 For Onboard Performance Tool. Not used by STAS.
 If this input is specified, the angle of taxiway input is required.
 (Optional) Runway width (same unit as runway distances)
 For Onboard Performance Tool. Not used by STAS.
 If this input is specified, the missed approach gradient
 requirement must be input.
 (Optional) Runway magnetic heading.
 For Onboard Performance Tool. Not used by STAS.
 If this input is specified, the runway width must be input.
 Note: Use a single space to separate RWYS and each of
 the optional inputs.

Line 2: The following data items in order:
 Runway designator (up to 7 characters enclosed in single quotes)
 Runway flag
 0 for TORA, TODA, ASDA
 1 for TORA, Clearway, Stopway
 TORA
 TODA or Clearway
 ASDA or Stopway
 Landing distance available (read, but not used by STAS)
 Slope for TOD and Flight Path (%)
 Slope for ASD (%)
 Slope for landing calculations (read, but not used by STAS)
 Number of Obstacles
 Runway surface type (Normal=0,Gravel=1,Grooved/Porous=2)
 (Grooved/Porous refers to Skid-Resistant per AC 150/5320-12C)

Line 3+: If the number of obstacles is greater than zero,
 include the following data items in order:

Height of obstacle number 1
 Distance to obstacle number 1
 Lateral offset of obstacle number 1
 Height of obstacle number 2
 Distance to obstacle number 2
 ... etc. ...

Example:

RWYS	180	2.5	150	275								
'01R'	1	8000	200	300	0	0.5	0.5	0.0	3	0		
30		1200	0									
38		2000	0									
55		2800	0									

RWYT Format: Turning departures

- Line 1: RWYT columns 2 through 5.
 (Optional) Angle of taxiway entry to runway in degrees.
 (Optional) Missed approach gradient requirement in percent.
 To be computed at actual missed approach speed.
 For Onboard Performance Tool. Not used by STAS.
 If this input is specified, the angle of taxiway input is required.
 (Optional) Runway width (same unit as runway distances)
 For Onboard Performance Tool. Not used by STAS.
 If this input is specified, the missed approach gradient requirement must be input.
 (Optional) Runway magnetic heading.
 For Onboard Performance Tool. Not used by STAS.
 If this input is specified, the runway width must be input.
 Note: Use a single space to separate RWYT and each of the optional inputs.
- Line 2: The following data items in order:
 Runway designator (up to 7 characters enclosed in single quotes)
 Runway flag
 0 for TORA, TODA, ASDA
 1 for TORA, Clearway, Stopway
 TORA
 TODA or Clearway
 ASDA or Stopway
 Landing distance available (read, but not used by STAS)
 Slope for TOD and Flight Path (%)
 Slope for ASD (%)
 Slope for landing calculations (read, but not used by STAS)
 Number of Obstacles
 Runway surface type (Normal=0, Gravel=1, Grooved/Porous=2)
 (Grooved/Porous refers to Skid-Resistant per AC 150/5320-12C)
- Line 3+: If the number of obstacles is greater than zero, include the following data items in order:
 Height of obstacle number 1
 Distance to obstacle number 1
 Lateral offset of obstacle number 1
 Height of obstacle number 2
 Distance to obstacle number 2
 Lateral offset of obstacle number 2
 ... etc. ...
- Last line: Turn data specification. FPTD(1) through FPTD(7) as defined in the SCAP Takeoff Specification.

Example:

RWYT	180	2.5	150	045							
'01R'	1	8000	200	300	0	0.5	0.5	0.0	3	0	
30		1200	0								
38		2000	0								
55		2800	0								

RWYU Format: Straight out departures.

Line 1: RWYU columns 2 through 5.

- (Optional) Angle of taxiway entry to runway in degrees.
- (Optional) Missed approach gradient requirement in percent.
To be computed at actual missed approach speed.
For Onboard Performance Tool. Not used by STAS.
If this input is specified, the angle of taxiway input is required.
- (Optional) Runway width (same unit as runway distances)
For Onboard Performance Tool. Not used by STAS.
If this input is specified, the missed approach gradient requirement must be input.
- (Optional) Runway magnetic heading.
For Onboard Performance Tool. Not used by STAS.
If this input is specified, the runway width must be input.

Note: Use a single space to separate RWYU and each of the optional inputs.

Line 2: The following data items in order:

- Runway designator (up to 7 characters enclosed in single quotes)
- Runway flag
 - 0 for TORA, TODA, ASDA
 - 1 for TORA, Clearway, Stopway
- TORA
- TODA or Clearway
- ASDA or Stopway
- Landing distance available (read, but not used by STAS)
- Slope for TOD and Flight Path (%)
- Slope for ASD (%)
- Slope for landing calculations (read, but not used by STAS)
- Number of Obstacles
- Runway surface type (Normal=0, Gravel=1, Grooved/Porous=2)
(Grooved/Porous refers to Skid-Resistant per AC 150/5320-12C)

Line 3+: If the number of obstacles is greater than zero,
include the following data items in order:

Height of obstacle number 1
Distance to obstacle number 1
Lateral offset of obstacle number 1
Height of obstacle number 2
Distance to obstacle number 2
Lateral offset of obstacle number 2
... etc. ...

Last line: Engine-out departure procedure
(up to 130 characters enclosed in single quotes)

Example:

```
RWYU 90 2.50 75 349
'35' 0 6201 6201 6201 6201 0.21 0.21 0.21 4 0
25 1294 0
35 1323 0
244 55457 0
406 68106 0
'USE STD SID, HOLD AT VOR/DME XYZ, AWAIT ATC INSTRUCTIONS'
```

RWYV Format: Turning departures

Line 1: RWYV columns 2 through 5.
 (Optional) Angle of taxiway entry to runway in degrees.
 (Optional) Missed approach gradient requirement in percent.
 To be computed at actual missed approach speed.
 For Onboard Performance Tool. Not used by STAS.
 If this input is specified, the angle of taxiway input is required.
 (Optional) Runway width (same unit as runway distances)
 For Onboard Performance Tool. Not used by STAS.
 If this input is specified, the missed approach gradient requirement must be input.
 (Optional) Runway magnetic heading.
 For Onboard Performance Tool. Not used by STAS.
 If this input is specified, the runway width must be input.
 Note: Use a single space to separate RWYV and each of the optional inputs.

Line 2: The following data items in order:
 Runway designator (up to 7 characters enclosed in single quotes)
 Runway flag
 0 for TORA, TODA, ASDA
 1 for TORA, Clearway, Stopway
 TORA
 TODA or Clearway
 ASDA or Stopway
 Landing distance available (read, but not used by STAS)
 Slope for TOD and Flight Path (%)
 Slope for ASD (%)
 Slope for landing calculations (read, but not used by STAS)
 Number of Obstacles
 Runway surface type (Normal=0, Gravel=1, Grooved/Porous=2)
 (Grooved/Porous refers to Skid-Resistant per AC 150/5320-12C)

Line 3+: If the number of obstacles is greater than zero, include the following data items in order:

 Height of obstacle number 1
 Distance to obstacle number 1
 Lateral offset of obstacle number 1

... etc. ...

Next line: Turn data specification. FPTD(1) through FPTD(7) as defined in the SCAP Takeoff Specification.

Last line: Engine-out departure procedure
 (up to 130 characters enclosed in single quotes)


```

RWYV 180 2.5 150 110
'01R' 1 8000 200 300 0 0.5 0.5 0.0 1 0
30 1200 0
2 5 1 1000 0 45 9.E20
'USE STD SID, HOLD AT VOR/DME XYZ, AWAIT ATC INSTRUCTIONS'

```

- ⇒ Note: In RWYU and RWYV, the line defining the engine-out procedure is needed even if no procedure is defined for the runway. In this situation, use one or more spaces enclosed with single quotes. All procedures should be less than 130 characters long.
- ⇒ Note: Stopway and Clearway distance inputs are limited to five characters, including the “-” sign if inputting a negative value.
- ⇒ Note: Intersection information can be added after the last line of each runway input. The Intersection input is NOT used by STAS, but can be entered as an optional input for OPT use. Below is the intersection input format:

After last line of each runway input, include the following data items in order (space delimited):

```

#INT      in columns 1 through 4
Intersection name (enclosed in single quotes)
Intersection offset from runway start
lineup angle
Slope for TOD and Flight path (%)
Slope for ASD (%)

```

Example:

```
RWYV 180 2.5 150
  '01R' 1 8000 200 300 0 0.5 0.5 0.0 1 0
    30 1200 0
    2 5 1 1000 0 45 9.E20
  'USE STD SID, HOLD AT VOR/DME XYZ, AWAIT ATC INSTRUCTIONS'
#INT 'A' 1200 90 0.6 0.6
```

Airport and Runway Comment Lines

STAS supports airport and runway comments in the STAS input file. Comment lines start with the letter 'H' in the first column. Airport comments are entered on the lines between the airport name and the runway definition. Runway comments follow the runway definition. Multiple lines are allowed, but must be consecutive.

Example:

```
AIRPORT2
ZXXX YANKEE DOODLE INTL SEATTLE, WA      FFBRBR      250
H ZXXX AIRPORT COMMENT LINE
RWYU 180 2.5 150
  '01R' 1 8000 200 300 0 0.5 0.5 0.0 1 0
    30 1200 0
  'USE STD SID, HOLD AT VOR/DME XYZ, AWAIT ATC INSTRUCTIONS'
H 01R RUNWAY COMMENT LINE
RWYU 180 2.5 150
  '17L' 1 8000 200 300 0 0.5 0.5 0.0 1 0
    30 1200 0
  'USE STD SID, HOLD AT VOR/DME XYZ, AWAIT ATC INSTRUCTIONS'
H THERE CAN BE MULTIPLE COMMENT LINES BUT
H THEY MUST BE CONSECUTIVE
```

1234567890123456789012345678901234567890123456789012345678901
10 20 30 40 50 60